# A Unified Outline of RSVP Research Projects

Flyxion

December 8, 2025

**Abstract**

This document presents a structured outline of the major ongoing research directions associated with the Relativistic Scalar–Vector Plenum (RSVP), together with closely aligned mathematical and philosophical projects that extend its scope. Each subsection focuses on a single initiative, emphasizing its motivation, conceptual foundations, mathematical structures, anticipated deliverables, and long-term research potential.

## 1 Introduction

The Relativistic Scalar–Vector Plenum (RSVP) is a field–theoretic framework that seeks to unify cosmological dynamics, cognition, semantic emergence, and physical law under a single lamphrodynamic ontology. Over the past several years, a network of related subprojects has emerged to formalize RSVP using derived algebraic geometry, shifted symplectic structures, categorical rewriting systems, and entropic field operators. The purpose of this essay is to provide a high–level but mathematically informed overview of these projects, clarifying their interrelations and outlining the conceptual map that connects them into a unified research program.

While many of these projects remain in active development, others have already reached intermediate states of formal articulation, including manuscripts, simulation infrastructures, and partial proofs. The following sections adopt a modular structure: each project is introduced by motivation and status, followed by a conceptual outline, mathematical formulation, and plans for near–term and long–term development.

# Part I
# RSVP Theoretical Frameworks

## 2 The Unified DAG–RSVP Framework

**Status:** Active.

The Unified DAG–RSVP framework aims to provide a mathematically rigorous description of RSVP using derived algebraic geometry (DAG), especially the theory of derived mapping stacks, shifted symplectic structures, and PTVV–type symplectic geometry. The central idea is that the scalar–vector fields $(\Phi, \vec{v}, S)$ should not be understood merely as smooth tensor fields on a manifold, but as sections of derived stacks equipped with $(-1)$–shifted symplectic forms, encoding the entropic, lamphrodynamic, or baryonic constraints that modulate field interactions.

From this perspective, entropic descent and lamphrodynamic flow become geometric flows on derived mapping spaces:

$$\mathcal{M} = \mathrm{Map}(X, \mathfrak{Plenum})$$

equipped with shifted cotangent complexes whose obstruction theory encodes precisely the "pinch points" associated with entropic smoothing. These singularities are not pathologies, but fundamental structural invariants of the RSVP process.

**Next Steps.** A formal AKSZ/BV–quantization construction appears possible by treating RSVP as a classical BV–theory whose gauge structure reflects entropy–preserving redundancies in the lamphrodynamic action functional. A future project involves a general quantization conjecture for RSVP fields using shifted symplectic methods.

## 3 Derived L–System Sigma Models

**Status:** Draft manuscript in development.

The project on Derived L–System Sigma Models attempts to reinterpret classical L–systems, traditionally used for biological morphogenesis and fractal generation, as formal sigma models valued in derived spaces of rewrite rules. The guiding intuition is that rewriting itself can be understood as a field evolution under derived symplectic constraints, so that grammar dynamics behave like a physical field theory equipped with BV–quantization.

Formally, one introduces a derived moduli space of rewriting rules,

$$\mathcal{R} = \mathrm{Map}(G, \mathfrak{Rules}),$$

and regards Lamphrodynamic action functionals as entropic gradients over this space. The resulting BV–differential describes propagating uncertainty and semantic change in the rewriting process, allowing "grammar evolution" to be viewed as a kind of continuum field flow, suitable for modeling both biological and cognitive emergence.

**Long–Term Goal.** Establish a mathematical equivalence between RSVP lamphrodynamics and a derived sigma–model theory of rewriting, thereby positioning L–systems as semantic "world–sheet" fields.

# 4 The RSVP–TARTAN Extension

**Status:** Ongoing development.

The TARTAN extension of RSVP introduces recursive tiling, Gray–code rewriting, and CRDT–style concurrency into RSVP field ontology. The purpose is to model how spatially local entropic interactions propagate across recursive micro–macro tiling hierarchies, preserving ethical or semantic constraints in the presence of concurrency.

The resulting tiling algebras define a geometric measure of semantic "ethical cost," modeled using Wasserstein geometry on the derived space of possible tilings. Furthermore, derived homotopical methods can be used to characterize tiling deformations, enabling a rigorous understanding of the semantic and cosmological flows underlying RSVP.

Future work includes integration with lamphrodynamic differential equations, GPU–accelerated simulation, and categorical formulations of CRDT concurrency.

# Part II

# Mathematical Infrastructure

## 5 SpherePop Calculus: Merge–Collapse Geometry

**Status:** Formal specification + Lean implementation in progress.

### 5.1 Motivation

SpherePop attempts to define computation geometrically rather than symbolically. Values ("agents") are represented as geometric balls—or more abstractly as sections of a bundle whose fibers are round metric spaces—and computation proceeds through two primitive operations:

1. *merge*: geometric union along a controlled boundary;
2. *collapse*: abstraction that eliminates internal structure while preserving boundary invariants.

The algebraic intuition is similar to monoidal categories of regions, but here the monoidal product is geometrically realized by union in a metric–sheaf topology.

### 5.2 Preliminary Formalism

Let $(X, d)$ be a metric space equipped with a sheaf of "population bundles" $\mathcal{P}$ whose sections $\sigma : U \to \mathcal{P}(U)$ represent spherical agents distributed over an open region $U \subset X$.

A *SpherePop state* is then a finite collection

$$S = \{\sigma_1, \ldots, \sigma_n\} \qquad \sigma_i \in \Gamma(U_i, \mathcal{P})$$

whose supports are pairwise disjoint or geometrically mergeable.

**Merge Operator.** Given two states $\sigma_i, \sigma_j$ defined on $U_i, U_j$, a merge is defined by

$$\mu : (\sigma_i, \sigma_j) \mapsto \sigma_{ij}$$

where $\sigma_{ij}$ is a section on $U_i \cup U_j$ satisfying

$$\sigma_{ij}|_{U_i} = \sigma_i, \qquad \sigma_{ij}|_{U_j} = \sigma_j$$

together with a boundary compatibility condition expressed as a pushout in the category of sheaves:

$$\mathcal{P}(U_i) \amalg_{\mathcal{P}(U_i \cap U_j)} \mathcal{P}(U_j).$$

**Collapse Operator.** Collapse is a geometric abstraction that replaces interior structure by a "collapsed" spherical profile. Formally, collapse is a functor

$$\kappa : \mathsf{Sh}(X) \to \mathsf{Sh}(X)$$

whose action on $\sigma$ forgets internal stratification while preserving a boundary measure:

$$\int_{\partial\sigma} \rho = \text{const}$$

where $\rho$ may represent information density or entropy flux.

## 5.3 Derived Interpretation

A major open conjecture is that SpherePop operations can be encoded as derived pushouts in a category of sheaf–valued stacks. In that case, collapse is a derived truncation functor (e.g., Postnikov tower) and merge becomes a homotopy colimit:

$$\sigma_{ij} \simeq \text{hocolim}\left(\sigma_i \leftarrow \sigma_i|_{U_i \cap U_j} \to \sigma_j\right).$$

This interpretation suggests that "computation" in SpherePop is simply the iterated formation of homotopy colimits, linking directly to semantic infrastructure and derived category theory.

## 5.4 Variational Form

We postulate an entropy–weighted functional measuring lamphrodynamic cost:

$$\mathcal{A}[\sigma] = \int_X \left(\alpha\|\nabla\sigma\|^2 + \beta S(\sigma)\right) dV$$

where the first term penalizes geometric fragmentation and the second term measures entropic potential. Collapse corresponds to steepest descent in $\mathcal{A}$ restricted to boundary–preserving homotopy classes.

# 6 Amplitwistor Algebra Scaffold

**Status:** conceptual scaffold + partial algebraic constructions.

## 6.1 Motivation

RSVP dynamics include lamphrodynamic flows that resemble scattering processes: information, curvature, and entropy exchange among localized quanta. Amplitwistor methods, originating in twistor theory and ambitwistor string models, offer a categorical route to define RSVP "scattering" in a shifted–symplectic geometric setting.

## 6.2 Mathematical Setup

Let $Z$ denote an ambitwistor space (complex sixfold in 4D space–time), and suppose RSVP fields define a map

$$\Phi : M \to Z.$$

Scattering amplitudes in twistor models arise as integrals over moduli of holomorphic curves in $Z$; by analogy, RSVP amplitudes may arise from derived mapping stacks of lamphrodynamic embeddings:

$$\mathrm{Map}(M, Z)_{\mathrm{hol}}.$$

The classical action takes a form reminiscent of ambitwistor string functionals:

$$S[\Phi] = \int_M \Phi^*(\omega)$$

where $\omega$ is a holomorphic symplectic form on $Z$.

## 6.3 Shifted Symplectic Structure

By PTVV theory, derived moduli spaces of maps into shifted symplectic targets inherit a natural $(-1)$–shifted symplectic structure. Thus, if $Z$ is shifted–symplectic, then the mapping space inherits:

$$\Omega_{\mathrm{RSVP}} \in \Omega^\bullet(\mathrm{Map}(M, Z))[-1].$$

**Conjecture.** RSVP amplitudes are (derived) integrals over Lagrangian intersections in $\mathrm{Map}(M, Z)$ analogous to Witten's twistor string construction but with entropy–weighting rather than conformal invariance.

# 7 Spectral Geometry and Cymatic Resonance

**Status:** mathematical chapter under active drafting.

## 7.1 Overview

A central aspect of RSVP physiology (both cosmological and cognitive) is the idea that lamphrodynamic fields exhibit spectral resonance, analogous to cortical oscillations or cosmological acoustic modes. Spectral geometry offers a rigorous framework for analyzing such resonances.

## 7.2 Laplace–Beltrami Formalism

Let $(M, g)$ be a Riemannian manifold with RSVP field variables $(\Phi, v, S)$. The Laplace–Beltrami operator acts on scalar functions and vector fields as:

$$\Delta_g \Phi = \nabla^a \nabla_a \Phi, \qquad \Delta_g v^b = \nabla_a \nabla^a v^b + R^b{}_a v^a.$$

Eigenmodes

$$\Delta_g \psi_k = \lambda_k \psi_k$$

describe resonant configurations which, under RSVP interpretation, correspond to preferred entropic or lamphrodynamic states.

## 7.3 Entropy–Weighted Spectral Measure

Define an entropy–weighted spectral measure

$$\rho(\lambda) = \sum_k e^{-S_k} \, \delta(\lambda - \lambda_k),$$

where $S_k$ measures entropic "cost" of mode $\psi_k$. This function behaves as a spectral density that "filters" high–entropy or low–entropy modes, suggesting a principled measure of spectral complexity useful for analyzing both brain dynamics and cosmological fluctuations.

## 7.4 Cymatic Interpretation

RSVP assumes that meaning–bearing oscillations correspond to eigenmodes of derived Laplacians on semantic manifolds. A working hypothesis is that *semantic resonance* can be modeled by the extremization of an entropy– modified spectral functional:

$$\mathcal{S}[\psi] = \int_M \Big( \|\nabla\psi\|^2 + \gamma\, \Phi[\psi] - \delta\, S[\psi] \Big)\, dV.$$

Critical points of $\mathcal{S}$ describe stationary states of lamphrodynamic cymatics.

# Part III

# Cosmology and Gravitational Interpretation

## 8 RSVP Cosmology

**Status:** Long-term monograph; major systems under development.

### 8.1 Motivation

Standard cosmology explains large-scale structure through expansion, dark energy, and inflationary processes. By contrast, RSVP proposes that the apparent expansion of the universe results from entropic smoothing of lamphrodynamic curvature fields, without requiring a literal metric expansion of space. The aim is to derive cosmological redshift from local entropy-flux geometry rather than global expansion of spatial sections.

### 8.2 Lamphrodynamic Stress–Energy

Let $(M, g)$ be a Lorentzian manifold and let $\Lambda_a$ denote a *lamphrodynamic current* combining entropy gradient and baryon flow:

$$\Lambda_a = \nabla_a S + \beta\, v_a,$$

where $S$ is entropy and $v^a$ a vector flow. A lamphrodynamic stress–energy tensor is conjectured to take the form:

$$T^{(\Lambda)}_{ab} = \alpha\, \Lambda_a \Lambda_b + \gamma\, (\nabla_a \Lambda_b + \nabla_b \Lambda_a) - \delta\, g_{ab}(\Lambda \cdot \Lambda),$$

so that gravitational curvature includes entropy–driven contributions:

$$R_{ab} - \tfrac{1}{2} R g_{ab} = T_{ab} + T^{(\Lambda)}_{ab}.$$

## 8.3 Entropic Raychaudhuri Equation

In general relativity, the Raychaudhuri equation governs the expansion of geodesic congruences. RSVP modifies the expansion scalar $\theta$ through entropy-flux coupling:

$$\frac{d\theta}{d\tau} = -\tfrac{1}{3}\theta^2 - \sigma_{ab}\sigma^{ab} + \omega_{ab}\omega^{ab} - R_{ab}u^a u^b + \mathcal{E},$$

where the entropic contribution $\mathcal{E}$ is proposed to be

$$\mathcal{E} = \lambda\,(u^a \nabla_a S) + \kappa\,(u^a v_a).$$

Negative contributions to $R_{ab}u^a u^b$ from $T^{(\Lambda)}_{ab}$ can mimic observable redshift without global metric expansion.

## 8.4 Cosmological Redshift Without Expansion

A provisional model treats photon frequency shift along null geodesics $\gamma$ as accumulated entropic drag:

$$z = \int_\gamma f(S, v)\, d\tau,$$

where the integrand is a scalar composed from $\Lambda_a$ contracted with the null tangent $k^a$:

$$f(S, v) = \eta\, \Lambda_a k^a.$$

This formulation preserves null geodesic structure while introducing redshift from entropic geometry rather than scale factor variation.

## 8.5 Lamphrodynamic Smoothing PDE

Assuming coupled evolution of $(\Phi, v, S)$ fields, a simplified cosmological equation takes the PDE form:

$$\partial_t \Phi - \nabla \cdot (v\Phi) = \mathcal{D}_\Phi \Delta\Phi - \xi S,$$

$$\partial_t v + (v \cdot \nabla)v = \mathcal{D}_v \Delta v - \nabla\Phi,$$

$$\partial_t S = \mathcal{D}_S \Delta S + \zeta\,\|\nabla v\|^2 - \chi\,\Phi^2.$$

Here $\xi, \zeta, \chi$ encode lamphrodynamic coupling. Spatial smoothing arises naturally as diffusion-type relaxation. Structure emerges from competition of $\Delta v$ and nonlinear $\Phi^2$ terms.

### 8.6 Perturbative Genesis

Early-universe perturbations can be modeled as solutions $\Phi = \Phi_0 + \delta\Phi$ where $\Phi_0$ is a nearly uniform lamphrodynamic background. Solving the linearized equations

$$\partial_t \delta\Phi = \mathcal{D}_\Phi \Delta \delta\Phi - \xi\, \delta S$$

$$\partial_t \delta S = \mathcal{D}_S \Delta \delta S$$

reveals that lamphrodynamic smoothing suppresses high-frequency modes while allowing long-wavelength structure to stabilize — recasting “inflation” as entropic equalization without metric expansion.

# 9 Gravity as Entropy Descent

**Status:** Comparative analysis + commentary article.

## 9.1 Motivation

Several modern approaches to gravity, notably Jacobson, Verlinde, and Carney, interpret gravitational dynamics as emergent from entropic information flow. RSVP proposes a distinct but related framework in which gravitational curvature is a lamphrodynamic effect of entropic gradients in $(\Phi, v, S)$ fields.

The difference is that RSVP derives gravitational geometry not from information-theoretic microstates at a boundary, but from lamphrodynamic interactions in the plenum itself.

## 9.2 Entropy–Curvature Coupling

Suppose curvature satisfies a modified Einstein equation:

$$R_{ab} - \tfrac{1}{2} R g_{ab} = 8\pi G\, T_{ab} + \Theta_{ab},$$

where $\Theta_{ab}$ is an entropy-induced correction. A natural choice is

$$\Theta_{ab} = \alpha\, \nabla_a \nabla_b S + \beta\, \nabla_{(a} v_{b)} + \gamma\, v_a v_b,$$

so that geodesic deviation incorporates entropy-flux directly.

## 9.3 Comparison with Jacobson

Jacobson derives Einstein equations from the Clausius relation

$$\delta Q = T \delta S$$

applied to local Rindler horizons. RSVP instead inserts $S$ as a dynamical field whose gradient contributes directly to curvature, leading to an internal, rather than horizon-based, source of geometric contribution. In RSVP, "temperature" becomes a lamphrodynamic function rather than a purely horizon-associated quantity.

## 9.4 Verlinde Entropic Gravity

Verlinde interprets gravity as emergent from holographic entropic elasticity. In RSVP, holography is not a necessary assumption; lamphrodynamics can exist without boundary thermodynamics. However, certain limits reproduce Verlinde-like entropic force laws when lamphrodynamic contributions dominate:

$$F \sim T\nabla S.$$

## 9.5 Carney Emergent Gravity

Carney's recent framework interprets gravitational fluctuations as emergent from entanglement and quantum information dynamics. RSVP differs in placing the source of curvature in classical lamphrodynamic fields $(\Phi, v, S)$ rather than quantum entanglement, although a possible connection emerges if RSVP fields are quantized via AKSZ/BV methods described earlier.

## 9.6 Phenomenological Outline

Observable consequences include:

1. redshift independent of cosmic expansion,
2. lamphrodynamic lensing without dark energy,
3. low-frequency smoothing of early universe perturbations,
4. entropy-induced acceleration of structure formation.

Whether these can match Planck–scale CMB spectra remains an open question for numerical experimentation.

# Part IV

# Exegetical and Editorial Projects

## 10 Appendix Series (A–O)

**Status:** Many sections well-developed; several under formulation.

### 10.1 Purpose

The Appendix series contains technical derivations, formal extensions, and philosophical commentaries intended to supplement the core monograph on RSVP cosmology and lamphrodynamics. Each appendix isolates a specific theme (e.g., spectral geometry, homotopy fibers, shifted symplectic quantization) and develops it independently, while maintaining coherence with the global RSVP project.

### 10.2 Structure

Each appendix is organized as a mini-monograph with the following elements:

1. background results from geometry and category theory,
2. formal definitions and conjectures,
3. derivations and intermediate lemmas,
4. proposed theorems or equivalence principles,
5. open questions and research directions.

### 10.3 Shifted Cotangent Stacks

A recurring structure across multiple appendices is the shifted cotangent stack $T^*[-1]\mathcal{M}$ associated with a mapping stack $\mathcal{M}$. Derived quantization leads naturally to BV operator constructions on $\mathcal{M}$, implying that RSVP fields may admit a quantized BV geometry.

Concretely, one may attempt to build lamphrodynamic functionals

$$S_{\mathrm{RSVP}} : T^*[-1]\mathcal{M} \to \mathbb{R}$$

whose derived critical locus encodes lamphrodynamic equilibria. This construction has not yet been carried out in detail, but forms a central long-term objective.

## 10.4 Homotopy Fibers and Singular Semantics

Several appendices investigate homotopy fibers of derived mapping stacks, particularly near singularities induced by entropic collapse. A crucial hypothesis is that semantic "pinch points"—points of high entropy compression—correspond precisely to derived critical loci of the BV action.

Tentative conjecture:

$$\mathrm{Crit}(S_{\mathrm{RSVP}}) \simeq \mathfrak{Plenum}\_\mathrm{Sing}$$

where the right-hand side denotes the locus of lamphrodynamic singularities.

## 10.5 Appendix O: Final Topic Selection

Appendix O remains provisional and will be chosen to address either:

1. a categorical unification of semantics with RSVP quantization, or
2. a deep comparison of lamphrodynamic cosmology with emergent gravity, or
3. a formal entropy–geometry theorem about redshift without expansion.

The decision will be made based on coherence with the completed monograph.

# Part V

# Narrative and Creative Projects

## 11 The Call from Ankyra

**Status:** Narrative outline; research screenplay.

### 11.1 Premise and Context

*The Call from Ankyra* is a narrative exploration of RSVP cosmology set in a far–future context in which Jupiter has become a star, named Ankyra. The story follows an exploratory mission to the Jovian moons and encounters a monolithic intelligence whose reasoning embodies a primitive version of RSVP.

### 11.2 Narrative Structure

1. *Act I:* unraveling the Ankyra event;
2. *Act II:* engagement with monolithic consciousness;
3. *Act III:* ethical lamphrodynamic choice facing humanity.

Each act introduces theoretical motifs drawn from RSVP, such as recursive entropy descent, lamphrodynamic inference, and semantic phase transitions.

### 11.3 Theoretical Embedding

The screenplay explicitly explores concepts of cosmic entropy smoothing, semantic cosmogenesis, and recursive lamphrodynamic agency. Dialogues embed mathematical metaphors, including derived mapping spaces and spectral cymatics, reflecting how cosmological understanding and ethical decision-making are intertwined in RSVP.

Possible extended motifs include the notion that apparent "expansion" is actually the semantic unfolding of entropy fields at cosmological scale.

# 12 Visions of a Spirit Seer

**Status:** Ongoing design and artistic development.

## 12.1 Overview

This visual project builds a modern creative interpretation of Emanuel Swedenborg's 1757 cosmological vision—famously described as the "last judgment" witnessed by a spirit seer. The year 1757 is treated not merely as a historical curiosity but as a symbolic moment of semantic phase transition in human cognition and cosmology.

## 12.2 Aesthetic and Geometry

Posters and visual compositions incorporate:

- Koenigsberg architecture,
- geometric typographic grids,
- spectral-inspired lighting,
- entropic and lamphrodynamic symbols.

Symbolic geometry is encouraged to illustrate derived symplectic flow, representing cosmic or cognitive singularities akin to RSVP "pinch points."

## 12.3 Mathematical Resonances

A speculative formal connection is that Swedenborg's cosmological vision may be interpreted metaphorically as a lamphrodynamic eigenmode: a "cymatic mode" representing semantic phase transition in collective cognition. Formally, such a mode might be modeled by an eigenfunction $\psi$ of an entropy-modified Laplacian, as described in Section **??**, though the present project leaves this interpretation metaphorical rather than technical.

# Part VI

# Infrastructure and Tooling

## 13 Versioning and Semantic Infrastructure

**Status:** Episodic; ongoing design and consolidation.

### 13.1 Purpose

Version control and semantic infrastructure ensure that theoretical manuscripts, simulation prototypes, and artistic modules remain organized, searchable, and interoperable under a growing research portfolio.

### 13.2 Semantic Versioning

A semantic versioning framework is being designed that encodes:

1. theoretical maturity,
2. mathematical rigor,
3. computational implementability,
4. narrative coherence,
5. research readiness.

Unlike ordinary semantic versioning (which tracks interface compatibility), this framework attempts to track the *semantic completeness* of research modules.

### 13.3 Long–Term Goal

Ultimately, semantic infrastructure should converge with SpherePop and DAG–RSVP into a unified *semantic computation platform* in which versions reflect structural progress across the research landscape rather than incremental code modifications.

# 14 Linearized Lamphrodynamic PDE: Existence and Uniqueness

We illustrate that, at least in a simplified linearized regime, the lamphrodynamic PDEs admit well-posed solutions.

Consider a scalar field $\phi(t,x)$ on the flat $d$-torus $T^d = (\mathbb{R}/2\pi\mathbb{Z})^d$ satisfying the linear equation

$$\partial_t \phi = \mathcal{D}\Delta\phi + F(t,x), \tag{1}$$

where $\mathcal{D} > 0$ is constant and $F$ is smooth (or suitably regular).

**Theorem 14.1** (Well-posedness of the Linearized Scalar Flow). *Let $\phi_0 \in H^s(T^d)$ for some $s \geq 0$ and $F \in L^2_{\mathrm{loc}}([0,\infty); H^s(T^d))$. Then there exists a unique solution*

$$\phi \in C([0,\infty); H^s(T^d))$$

*to equation (1) with initial data $\phi(0,x) = \phi_0(x)$. Moreover, the solution depends continuously on the initial data and forcing.*

*Proof (sketch).* On the torus, $\Delta$ has a complete orthonormal basis of eigenfunctions $e_k(x) = e^{ik\cdot x}$ with eigenvalues $-|k|^2$, $k \in \mathbb{Z}^d$. Expand

$$\phi(t,x) = \sum_k \hat{\phi}_k(t) e_k(x), \quad F(t,x) = \sum_k \hat{F}_k(t) e_k(x).$$

Equation (1) reduces to an infinite family of ODEs

$$\partial_t \hat{\phi}_k(t) = -\mathcal{D}|k|^2 \hat{\phi}_k(t) + \hat{F}_k(t),$$

each solved by variation of constants:

$$\hat{\phi}_k(t) = e^{-\mathcal{D}|k|^2 t}\hat{\phi}_k(0) + \int_0^t e^{-\mathcal{D}|k|^2(t-\tau)} \hat{F}_k(\tau)\, d\tau.$$

Regularity and convergence in $H^s$ follow from standard estimates on the heat semigroup $e^{t\mathcal{D}\Delta}$ and the assumed regularity of $F$ and $\phi_0$. Uniqueness follows from linearity and Grönwall-type estimates. Continuity with respect to data is immediate from the explicit representation. $\square$

*Remark* 14.2. Nonlinear lamphrodynamic systems (with advective terms like $(v \cdot \nabla)\phi$ and nonlinear couplings) require more sophisticated PDE tools, but the linearized case illustrates that the RSVP PDE framework fits within standard well-posedness theory on compact manifolds.

# 15 A Simple Entropy–Curvature Bookkeeping Identity

We now justify, at least algebraically, the curvature decomposition used in the "Gravity as Entropy Descent" section.

**Proposition 15.1** (Curvature Decomposition Identity)**.** *Let $(M, g)$ be a Lorentzian manifold and suppose the Einstein equations are modified to*

$$R_{ab} - \tfrac{1}{2}Rg_{ab} = 8\pi G\, T_{ab}^{\text{matter}} + \Theta_{ab}, \tag{2}$$

*where $\Theta_{ab}$ is a symmetric tensor built from lamphrodynamic fields $(\Phi, v, S)$ as in the main text. Then the Ricci tensor $R_{ab}$ admits a decomposition*

$$R_{ab} = R_{ab}^{\text{matter}} + R_{ab}^{\Lambda},$$

*where $R_{ab}^{\text{matter}}$ depends only on $T_{ab}^{\text{matter}}$ and $R_{ab}^{\Lambda}$ depends only on $\Theta_{ab}$ and $g_{ab}$.*

*Proof.* Start from (2) and denote $T_{ab}^{\text{tot}} = T_{ab}^{\text{matter}} + \Theta_{ab}/(8\pi G)$. Then

$$R_{ab} - \tfrac{1}{2}Rg_{ab} = 8\pi G\, T_{ab}^{\text{tot}}.$$

Take the trace with $g^{ab}$:

$$R - 2R = 8\pi G\, T^{\text{tot}}, \qquad -R = 8\pi G\, T^{\text{tot}},$$

so $R = -8\pi G\, T^{\text{tot}}$. Substituting back,

$$R_{ab} + 4\pi G\, T^{\text{tot}} g_{ab} = 8\pi G\, T_{ab}^{\text{tot}},$$

or

$$R_{ab} = 8\pi G \left( T_{ab}^{\text{tot}} - \tfrac{1}{2}T^{\text{tot}} g_{ab} \right).$$

Now decompose $T_{ab}^{\text{tot}} = T_{ab}^{\text{matter}} + \Theta_{ab}/(8\pi G)$. Then

$$R_{ab} = 8\pi G \left( T_{ab}^{\text{matter}} - \tfrac{1}{2}T^{\text{matter}} g_{ab} \right) + \left( \Theta_{ab} - \tfrac{1}{2}\Theta g_{ab} \right).$$

Define

$$R_{ab}^{\text{matter}} := 8\pi G \left( T_{ab}^{\text{matter}} - \tfrac{1}{2}T^{\text{matter}} g_{ab} \right), \qquad R_{ab}^{\Lambda} := \Theta_{ab} - \tfrac{1}{2}\Theta g_{ab}.$$

Then $R_{ab} = R_{ab}^{\text{matter}} + R_{ab}^{\Lambda}$ as claimed. □

*Remark* 15.2. Proposition 15.1 is purely algebraic and holds for any symmetric $\Theta_{ab}$. RSVP's specific contribution is to propose particular lamphrodynamic forms for $\Theta_{ab}$ (e.g. involving $\nabla_a \nabla_b S$ and $v_a v_b$) and then interpret $R^{\Lambda}_{ab}$ as encoding entropy descent in the geometry.

## 16 Energy Estimates for a Coupled $\Phi$–$S$ System

We now consider a simplified coupled lamphrodynamic system on the flat $d$–dimensional torus $T^d = (\mathbb{R}/2\pi\mathbb{Z})^d$, capturing the interaction between a scalar potential $\Phi$ and an entropy field $S$. A toy version of the RSVP cosmological PDEs reads:

$$\partial_t \Phi = \mathcal{D}_\Phi \Delta \Phi - \xi S, \tag{3}$$

$$\partial_t S = \mathcal{D}_S \Delta S - \chi \Phi^2, \tag{4}$$

where $\mathcal{D}_\Phi, \mathcal{D}_S > 0$ are diffusion constants and $\xi, \chi$ are real coupling parameters.

We seek *a priori* energy estimates for $(\Phi, S)$ in $L^2$–based Sobolev spaces. The following result is illustrative rather than exhaustive; it demonstrates how standard PDE methods can be applied to lamphrodynamic systems.

**Theorem 16.1** (Basic Energy Inequality for the Coupled System)**.** *Let* $(\Phi, S)$ *solve* (7)–(8) *on* $[0, T] \times T^d$ *with initial data*

$$\Phi(0, \cdot) = \Phi_0 \in L^2(T^d), \qquad S(0, \cdot) = S_0 \in L^2(T^d).$$

*Define the energy functional*

$$E(t) = \frac{1}{2}\|\Phi(t)\|_{L^2}^2 + \frac{\alpha}{2}\|S(t)\|_{L^2}^2$$

*for some constant* $\alpha > 0$*. Then there exist constants* $C_1, C_2 \geq 0$ *(depending on the parameters and the dimension) such that*

$$\frac{d}{dt}E(t) \leq -C_1\Big(\|\nabla\Phi(t)\|_{L^2}^2 + \|\nabla S(t)\|_{L^2}^2\Big) + C_2 E(t)^2. \tag{5}$$

*In particular, for sufficiently small initial data (in* $L^2$ *norm),* $E(t)$ *remains bounded for* $t \in [0, T]$*.*

*Proof (sketch).* Take the $L^2$ inner product of (7) with $\Phi$ and integrate over $T^d$:

$$\frac{1}{2}\frac{d}{dt}\|\Phi\|_{L^2}^2 = \int_{T^d} \Phi \partial_t \Phi \, dx = \mathcal{D}_\Phi \int \Phi \Delta \Phi \, dx - \xi \int \Phi S \, dx.$$

By integration by parts and periodicity,

$$\int \Phi \Delta \Phi \, dx = -\|\nabla \Phi\|_{L^2}^2.$$

Thus

$$\frac{1}{2}\frac{d}{dt}\|\Phi\|_{L^2}^2 = -\mathcal{D}_\Phi\|\nabla\Phi\|_{L^2}^2 - \xi\int \Phi S\,dx. \tag{6}$$

Similarly, multiply (8) by $\alpha S$ and integrate:

$$\frac{\alpha}{2}\frac{d}{dt}\|S\|_{L^2}^2 = \alpha\mathcal{D}_S\int S\Delta S\,dx - \alpha\chi\int S\Phi^2\,dx = -\alpha\mathcal{D}_S\|\nabla S\|_{L^2}^2 - \alpha\chi\int S\Phi^2\,dx.$$

Adding this to (10) gives

$$\frac{d}{dt}E(t) = -\mathcal{D}_\Phi\|\nabla\Phi\|_{L^2}^2 - \alpha\mathcal{D}_S\|\nabla S\|_{L^2}^2 - \xi\int \Phi S\,dx - \alpha\chi\int S\Phi^2\,dx.$$

We bound the mixed terms using Hölder and Young inequalities. First,

$$\left|\int \Phi S\,dx\right| \le \|\Phi\|_{L^2}\|S\|_{L^2} \le \frac{1}{2\epsilon_1}\|\Phi\|_{L^2}^2 + \frac{\epsilon_1}{2}\|S\|_{L^2}^2$$

for any $\epsilon_1 > 0$. Next, for the cubic term we use Gagliardo–Nirenberg plus Young; schematically,

$$\left|\int S\Phi^2\,dx\right| \le \|S\|_{L^2}\|\Phi\|_{L^4}^2 \le C\|S\|_{L^2}\|\Phi\|_{L^2}^{2(1-\theta)}\|\nabla\Phi\|_{L^2}^{2\theta}$$

for some $\theta \in (0,1)$ (depending on dimension). Another Young splitting then yields

$$\left|\int S\Phi^2\,dx\right| \le \epsilon_2\|\nabla\Phi\|_{L^2}^2 + C(\epsilon_2)\|S\|_{L^2}^p\|\Phi\|_{L^2}^q$$

for suitable exponents $p, q$ and constant $C(\epsilon_2)$.

Collecting terms and absorbing the gradient contributions into the negative definite part, we obtain an inequality of the form

$$\frac{d}{dt}E(t) \le -C_1(\mathcal{D}_\Phi, \mathcal{D}_S, \epsilon_2)\Big(\|\nabla\Phi\|_{L^2}^2 + \|\nabla S\|_{L^2}^2\Big) + C_2E(t)^2,$$

where we estimated $\|S\|_{L^2}^p\|\Phi\|_{L^2}^q \lesssim E(t)^2$ for some universal constant $C_2$ (depending on the norms and exponents). The desired inequality (9) follows. □

*Remark* 16.2. For small initial energy $E(0)$, a standard differential inequality argument (Grönwall plus a logistic-type bound) implies that $E(t)$ remains bounded for finite times. This basic mechanism underlies the expectation that lamphrodynamic systems are dynamically "tamed" by diffusion, even in the presence of nonlinear couplings.

# 17 Categorical Structure of Merge–Collapse as Homotopy Colimit

We now sketch a categorical viewpoint on the SpherePop merge–collapse operations, connecting them to homotopy colimits in a suitable model category of geometric configurations.

## 17.1 Configuration Category

Let $\mathcal{C}$ denote a category whose objects are finite configurations of labeled balls (spheres) in a manifold $X$ (or, more abstractly, sections of a population sheaf $\mathcal{P}$ over open subsets of $X$). Morphisms are generated by:

- embeddings that relabel or reposition spheres without changing their intrinsic state;
- *merge* morphisms identifying two overlapping or adjacent spheres into a single object;
- *collapse* morphisms passing from a structured configuration to a more abstract one preserving boundary invariants.

We envisage a model category structure on a suitable enlargement of $\mathcal{C}$ (e.g. simplicial presheaves or topological diagrams) in which merge morphisms are cofibrations and weak equivalences encode homotopy equivalence of configurations.

**Definition 17.1.** A *SpherePop diagram* is a small diagram $D : I \to \mathcal{C}$ indexed by a finite category $I$, whose shapes capture the combinatorial structure of a computation (sequence of merges and collapses).

## 17.2 Homotopy Colimit Viewpoint

Given a diagram $D : I \to \mathcal{C}$, we can form its ordinary colimit $\operatorname{colim} D$ (if it exists) and, in a model–categorical setting, its homotopy colimit $\operatorname{hocolim} D$.

**Lemma 17.2** (Merge as Pushout)**.** *Consider a configuration where two spheres $A$ and $B$ overlap along a common boundary region $A \cap B$. The* merge *configuration $A \cup B$ can be realized as a pushout in $\mathcal{C}$ of the diagram*

$$A \leftarrow A \cap B \to B.$$

*Proof.* By construction, $A \cup B$ is obtained by gluing $A$ and $B$ along their common intersection, identifying points in $A \cap B$. In any category of spaces (or sheaves) where such gluings are defined, this is precisely the universal object with maps from $A$ and $B$ that agree on $A \cap B$. Thus $A \cup B$ is a pushout. □

To understand *homotopy* pushouts, we assume $\mathcal{C}$ carries a model structure (for instance, as a full subcategory of a simplicial presheaf model category) such that inclusions of sub-configurations are cofibrations.

**Proposition 17.3** (Merge as Homotopy Colimit)**.** *Suppose the inclusions $A \cap B \hookrightarrow A$ and $A \cap B \hookrightarrow B$ are cofibrations in the model category containing $\mathcal{C}$. Then the pushout configuration $A \cup B$ is a homotopy pushout of the diagram $A \leftarrow A \cap B \rightarrow B$, and hence computes the homotopy colimit* $\operatorname{hocolim}(A \leftarrow A \cap B \rightarrow B)$.

*Proof (sketch).* In a cofibrantly generated model category, any pushout along a cofibration preserves homotopy types in the sense that the canonical map from the homotopy pushout to the strict pushout is a weak equivalence. Explicitly, if $f, g$ are cofibrations, then

$$\operatorname{hocolim}(A \leftarrow A \cap B \rightarrow B) \xrightarrow{\simeq} A \cup B$$

is a weak equivalence. The hypothesis that $A \cap B \rightarrow A$ and $A \cap B \rightarrow B$ are cofibrations ensures that the mapping cylinder or simplicial resolutions used to construct the homotopy pushout do not alter the homotopy type. This is a standard result in model category theory (see, e.g., Quillen or Hovey). Thus, up to weak equivalence, $A \cup B$ computes the homotopy colimit. □

*Remark* 17.4. The proposition expresses merge as a homotopy colimit operation. Intuitively, SpherePop "computation" can be regarded as building increasingly complex configurations via iterated homotopy colimits of basic spheres. Collapse can then be interpreted as a functor from $\mathcal{C}$ to its homotopy category (or a suitable localization) that contracts internal homotopy–equivalent subdiagrams while preserving boundary data.

## 17.3 Toward a Universal Property of Collapse

A fully rigorous account of collapse requires specifying a localization $\mathcal{C}[W^{-1}]$ with respect to a class $W$ of weak equivalences that encode "internally invisible" structure (e.g. contractible clusters of spheres). One may then view collapse as the composition

$$\kappa : \mathcal{C} \longrightarrow \mathcal{C}[W^{-1}] \longrightarrow \mathcal{C}_\partial,$$

where $\mathcal{C}_\partial$ retains only boundary data and coarse homotopy type.

**Conjecture 17.5** (Universal Collapse)**.** *There exists a localization and boundary category $\mathcal{C}_\partial$ such that for each configuration $X$, the collapsed object $\kappa(X)$ is a terminal object among all objects $Y$ equipped with a map $X \rightarrow Y$ that is:*

1. *a weak equivalence on homotopy type of internal regions, and*
2. *an isomorphism on homotopy type of boundary data.*

If true, this would justify the intuition that collapse is the "coarsest" configuration preserving boundary semantics, thus aligning precisely with the SpherePop idea of abstraction by eliminating internal complexity while preserving externally visible invariants.

# 18 Energy Estimates for a Coupled $\Phi$–$S$ System

We now consider a simplified coupled lamphrodynamic system on the flat $d$–dimensional torus $T^d = (\mathbb{R}/2\pi\mathbb{Z})^d$, capturing the interaction between a scalar potential $\Phi$ and an entropy field $S$. A toy version of the RSVP cosmological PDEs reads:

$$\partial_t \Phi = \mathcal{D}_\Phi \Delta \Phi - \xi S, \tag{7}$$

$$\partial_t S = \mathcal{D}_S \Delta S - \chi \Phi^2, \tag{8}$$

where $\mathcal{D}_\Phi, \mathcal{D}_S > 0$ are diffusion constants and $\xi, \chi$ are real coupling parameters.

We seek *a priori* energy estimates for $(\Phi, S)$ in $L^2$–based Sobolev spaces. The following result is illustrative rather than exhaustive; it demonstrates how standard PDE methods can be applied to lamphrodynamic systems.

**Theorem 18.1** (Basic Energy Inequality for the Coupled System)**.** *Let* $(\Phi, S)$ *solve* (7)–(8) *on* $[0,T] \times T^d$ *with initial data*

$$\Phi(0,\cdot) = \Phi_0 \in L^2(T^d), \qquad S(0,\cdot) = S_0 \in L^2(T^d).$$

*Define the energy functional*

$$E(t) = \frac{1}{2}\|\Phi(t)\|_{L^2}^2 + \frac{\alpha}{2}\|S(t)\|_{L^2}^2$$

*for some constant* $\alpha > 0$*. Then there exist constants* $C_1, C_2 \geq 0$ *(depending on the parameters and the dimension) such that*

$$\frac{d}{dt}E(t) \leq -C_1\Big(\|\nabla\Phi(t)\|_{L^2}^2 + \|\nabla S(t)\|_{L^2}^2\Big) + C_2 E(t)^2. \tag{9}$$

*In particular, for sufficiently small initial data (in* $L^2$ *norm),* $E(t)$ *remains bounded for* $t \in [0,T]$*.*

*Proof (sketch).* Take the $L^2$ inner product of (7) with $\Phi$ and integrate over $T^d$:

$$\frac{1}{2}\frac{d}{dt}\|\Phi\|_{L^2}^2 = \int_{T^d} \Phi \partial_t \Phi \, dx = \mathcal{D}_\Phi \int \Phi \Delta \Phi \, dx - \xi \int \Phi S \, dx.$$

By integration by parts and periodicity,

$$\int \Phi \Delta \Phi \, dx = -\|\nabla \Phi\|_{L^2}^2.$$

Thus

$$\frac{1}{2}\frac{d}{dt}\|\Phi\|_{L^2}^2 = -\mathcal{D}_\Phi\|\nabla\Phi\|_{L^2}^2 - \xi\int \Phi S\,dx. \tag{10}$$

Similarly, multiply (8) by $\alpha S$ and integrate:

$$\frac{\alpha}{2}\frac{d}{dt}\|S\|_{L^2}^2 = \alpha\mathcal{D}_S\int S\Delta S\,dx - \alpha\chi\int S\Phi^2\,dx = -\alpha\mathcal{D}_S\|\nabla S\|_{L^2}^2 - \alpha\chi\int S\Phi^2\,dx.$$

Adding this to (10) gives

$$\frac{d}{dt}E(t) = -\mathcal{D}_\Phi\|\nabla\Phi\|_{L^2}^2 - \alpha\mathcal{D}_S\|\nabla S\|_{L^2}^2 - \xi\int \Phi S\,dx - \alpha\chi\int S\Phi^2\,dx.$$

We bound the mixed terms using Hölder and Young inequalities. First,

$$\left|\int \Phi S\,dx\right| \le \|\Phi\|_{L^2}\|S\|_{L^2} \le \frac{1}{2\epsilon_1}\|\Phi\|_{L^2}^2 + \frac{\epsilon_1}{2}\|S\|_{L^2}^2$$

for any $\epsilon_1 > 0$. Next, for the cubic term we use Gagliardo–Nirenberg plus Young; schematically,

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2. *an isomorphism on homotopy type of boundary data.*

If true, this would justify the intuition that collapse is the "coarsest" configuration preserving boundary semantics, thus aligning precisely with the SpherePop idea of abstraction by eliminating internal complexity while preserving externally visible invariants.

# 20 From SpherePop Configurations to Derived Stacks

We now outline a speculative but conceptually coherent interpretation of SpherePop configurations inside a derived geometric framework. While the following discussion is intentionally informal, it provides a plausible route toward unifying SpherePop, DAG, and RSVP at a structural level.

## 20.1 Piecewise–Smooth Stratified Spaces

Let $\mathsf{Strat}_X$ denote the category of stratified spaces embedded in a smooth ambient manifold $X$. Each SpherePop configuration may be regarded as a finite union of stratified subspaces:

$$\sigma = \bigcup_{i=1}^{n} \overline{B_i} \subset X,$$

where each $B_i$ is an open ball (or more generally a smoothly embedded region carrying additional data). Stratifications arise naturally at intersections $B_i \cap B_j$ and along collapse boundaries.

**Definition 20.1.** The *SpherePop moduli functor* $\mathcal{M}_{\mathrm{SP}}$ assigns to each test space $U$ the groupoid (or $\infty$–groupoid) of SpherePop configurations parametrized by $U$. Informally,

$$\mathcal{M}_{\mathrm{SP}}(U) := \{\sigma \subset X \times U\}_{/\sim}$$

modulo isotopy or boundary–preserving equivalence.

The idea is that $\mathcal{M}_{\mathrm{SP}}$ packages all possible SpherePop states as a single moduli object. One may impose additional structure (entropy, mode labels, etc.) to obtain enriched versions.

## 20.2 Toward Derived Moduli

Derived algebraic geometry provides tools to study moduli problems with singularities, obstructions, and higher automorphisms. Since merge and collapse operations create gluings and identifications, the moduli of SpherePop configurations is naturally plagued by singularities that arise from intersections. Derived methods systematically capture such defects.

Informally, one may conjecture the existence of a derived enhancement

$$\mathbb{R}\mathcal{M}_{\mathrm{SP}}$$

of $\mathcal{M}_{\mathrm{SP}}$ incorporating derived intersections of stratified regions, leading to a derived cotangent complex whose cohomology governs infinitesimal merge/collapse deformations and obstructions.

## 20.3 Homotopy Colimits Inside a Derived Moduli Stack

Earlier we observed that merge operations realize homotopy colimits of the diagram $A \leftarrow A \cap B \rightarrow B$ in a model category setting. Inside a derived moduli context, this suggests that merge corresponds to a derived pushout in $\mathbb{R}\mathcal{M}_{\mathrm{SP}}$:

$$\sigma_{AB} \simeq \mathrm{hocolim}_{\mathbb{R}\mathcal{M}_{\mathrm{SP}}}\Big(A \leftarrow A \cap B \rightarrow B\Big).$$

More precisely, the pushout lives in the $\infty$–category of derived stacks, and the standard DAG result that pushouts along derived closed immersions compute homotopy pushouts in the $\infty$–topos suggests that the SpherePop merge is (up to homotopy) already a derived geometric gluing.

**Lemma 20.2** (Derived Gluing Heuristic)**.** *Assuming the inclusions $A \cap B \hookrightarrow A$, $A \cap B \hookrightarrow B$ are derived closed immersions in the sense of DAG, the pushout $A \cup B$ is computed (up to higher homotopy) by the $\infty$–categorical colimit in $\mathbb{R}\mathcal{M}_{\mathrm{SP}}$.*

*Proof (heuristic sketch).* In the setting of derived stacks, colimits are homotopy colimits by construction. If the inclusions are derived closed immersions (or cofibrations in an $\infty$–categorical sense), then the strict gluing along $A \cap B$ coincides with the derived gluing, up to equivalence. See PTVV and Toën–Vezzosi for parallel constructions in the derived category of schemes and Artin stacks. □

## 20.4 Collapse and Derived Truncation

Collapse loses internal homotopical information. In DAG, truncation functors $\tau_{\leq n}$ play the role of forgetting higher homotopy in a controlled way.

**Conjecture 20.3** (Derived Collapse)**.** *The SpherePop collapse operator $\kappa$ admits a derived interpretation as a functor*

$$\kappa : \mathbb{R}\mathcal{M}_{\mathrm{SP}} \longrightarrow \tau_{\leq 0}\mathbb{R}\mathcal{M}_{\mathrm{SP}}$$

*preserving boundary strata and coarse homotopy type while discarding higher homotopical structure internal to merged regions.*

In this view, collapse corresponds to passing from a derived moduli stack to its 0–truncation, i.e. taking the "coarse moduli" of a configuration while preserving boundary geometry. This precisely captures the intuition that SpherePop computation acts by homotopical merge followed by semantic abstraction.

## 20.5 Relation to RSVP and Shifted Symplectic Structures

Finally, RSVP's lamphrodynamic fields $(\Phi, v, S)$ live naturally in a (shifted) symplectic setting. If $\mathbb{R}\mathcal{M}_{\text{SP}}$ can be shown to admit a shifted symplectic structure, then SpherePop configurations could be viewed as classical lamphrodynamic states in a derived phase space, parallel to the DAG–RSVP framework.

This suggests a deeper derived equivalence:

$$\text{RSVP} \simeq \text{Lamphrodynamic Structure on } \mathbb{R}\mathcal{M}_{\text{SP}}$$

in an appropriate derived category. Establishing such an equivalence would rigorously integrate geometric computation with lamphrodynamic physics.

## 21 Shifted Cotangent Stacks and Lamphrodynamic Phase Space

We now describe how a shifted cotangent stack associated with the SpherePop moduli may be interpreted as a lamphrodynamic phase space, in parallel with the DAG–RSVP framework.

Let $\mathcal{M}_{\mathrm{SP}}$ be the (derived) moduli stack of SpherePop configurations as discussed above, and assume we have constructed a derived enhancement $\mathbb{R}\mathcal{M}_{\mathrm{SP}}$. Following PTVV, we may consider the shifted cotangent stack

$$T^*[n]\Big(\mathbb{R}\mathcal{M}_{\mathrm{SP}}\Big),$$

for some integer shift $n$, as a candidate phase space.

**Definition 21.1.** A *lamphrodynamic phase space* for SpherePop is a shifted cotangent stack $T^*[k]\Big(\mathbb{R}\mathcal{M}_{\mathrm{SP}}\Big)$ equipped with a $k$–shifted symplectic form $\omega_{\mathrm{SP}}$ in the sense of PTVV.

In the RSVP context, one expects $k = -1$ or $k = 0$ to be the most natural: $k = -1$ corresponds to BV–type theories, while $k = 0$ corresponds to classical symplectic geometry.

**Conjecture 21.2** (Shifted Symplectic Structure on SpherePop Moduli)**.** *There exists a shifted symplectic structure*

$$\omega_{\mathrm{SP}} \in \mathcal{A}^{2,cl}\Big(T^*[-1]\Big(\mathbb{R}\mathcal{M}_{\mathrm{SP}}\Big)\Big)$$

*obtained by transgressing a universal entropic 2-form along the universal SpherePop family.*

Formally establishing this would align SpherePop computation with the standard DAG quantization paradigm, allowing AKSZ/BV methods to be applied to lamphrodynamic computation.

## 22 AKSZ/BV Action for Lamphrodynamic Fields

We turn to an AKSZ-inspired formulation of lamphrodynamic RSVP fields.

Let $(\Sigma, \eta)$ be a source supermanifold (or dg-manifold) encoding spacetime plus degree shifts, and let $(\mathcal{X}, \omega_{\mathcal{X}})$ be a target shifted symplectic derived manifold that parameterizes RSVP field configurations (e.g. including $(\Phi, v, S)$ and their conjugate momenta or antifields). The AKSZ construction defines an action functional on the mapping space

$$\mathrm{Map}(\Sigma, \mathcal{X}).$$

**Definition 22.1** (Lamphrodynamic AKSZ-type Action)**.** Assume $\mathcal{X}$ is endowed with a $(n-1)$–shifted symplectic form $\omega_{\mathcal{X}}$ and a Hamiltonian function $\Theta$ of degree $n$ such that $\{\Theta, \Theta\} = 0$ with respect to the induced Poisson bracket. Then the lamphrodynamic AKSZ action is defined as

$$S_{\mathrm{AKSZ}}[\Phi] = \int_{\Sigma} \langle \Phi^*(\alpha_{\mathcal{X}}), d\Phi \rangle + \Phi^*(\Theta),$$

where $\alpha_{\mathcal{X}}$ is a potential for $\omega_{\mathcal{X}}$ and $\Phi : \Sigma \to \mathcal{X}$ ranges over maps in some appropriate dg-category.

The condition $\{\Theta, \Theta\} = 0$ guarantees that $S_{\mathrm{AKSZ}}$ satisfies the classical master equation in the BV sense, making it a consistent lamphrodynamic field theory candidate. In RSVP, one would encode entropic gradients, vector flows, and scalar potentials in the data of $\mathcal{X}$ and $\Theta$, with $\Theta$ capturing lamphrodynamic energy/entropy production.

# 23 A Prototype SpherePop Lagrangian

We now propose a more concrete Lagrangian for a continuum SpherePop field $\sigma(x)$ coupled to an entropy field $S(x)$ on a Riemannian manifold $(X, g)$.

**Definition 23.1** (SpherePop Lagrangian Density)**.** Let $\sigma : X \to \mathbb{R}_{\geq 0}$ denote a "density of spheres" and $S : X \to \mathbb{R}$ an entropy field. Define the Lagrangian density

$$\mathcal{L}_{\mathrm{SP}}(\sigma, S, \nabla\sigma, \nabla S) = \frac{\alpha}{2}\|\nabla\sigma\|^2 + \frac{\beta}{2}\|\nabla S\|^2 + V(\sigma, S),$$

where $V(\sigma, S)$ is a potential encoding merge/collapse preferences and lamphrodynamic coupling.

The associated action is

$$\mathcal{A}_{\mathrm{SP}}[\sigma, S] = \int_X \mathcal{L}_{\mathrm{SP}}\, dV_g.$$

The Euler–Lagrange equations formally read:

$$-\alpha\Delta\sigma + \partial_\sigma V(\sigma, S) = 0, \qquad -\beta\Delta S + \partial_S V(\sigma, S) = 0.$$

**Example 23.2.** A typical choice of potential might be

$$V(\sigma, S) = \lambda_1\sigma^2 + \lambda_2\sigma S + \lambda_3 e^{-S},$$

where the last term penalizes low-entropy configurations and encodes an explicit bias toward entropy increase. The mixed term $\lambda_2\sigma S$ couples sphere density to entropy.

In a fully derived context, $\sigma$ and $S$ would be replaced by sections into derived stacks, and the action would be lifted to a BV functional on a shifted cotangent stack as outlined above.

# 24 Merge as Homotopy Colimit in an $\infty$–Topos

We sharpen the justification of merge as a homotopy colimit by situating SpherePop configurations in an $\infty$–topos framework.

Let $\mathcal{E}$ be an $\infty$–topos (e.g. an $\infty$–category of sheaves of spaces on a site associated to $X$). Let $\mathcal{O}$ denote the full sub-$\infty$–category of $\mathcal{E}$ consisting of objects that encode SpherePop configurations (e.g. stratified sheaves, constructible sheaves, or similar).

**Definition 24.1.** A *merge diagram* in $\mathcal{O}$ is a span

$$A \xleftarrow{i} A \cap B \xrightarrow{j} B$$

where $i$ and $j$ are monomorphisms (inclusions) in $\mathcal{E}$ representing a common subconfiguration.

**Theorem 24.2** (Merge as Homotopy Pushout in an $\infty$–Topos)**.** *In an $\infty$–topos $\mathcal{E}$, every pushout square is automatically a homotopy pushout square. In particular, the merged configuration $A \cup B := A \amalg_{A\cap B} B$ computes the homotopy colimit*

$$A \cup B \simeq \operatorname{hocolim}\Big(A \leftarrow A \cap B \rightarrow B\Big),$$

*in the $\infty$–categorical sense.*

*Proof (sketch).* By the axioms of an $\infty$–topos, colimits are *homotopy invariant*: the $\infty$–category $\mathcal{E}$ is presentable and is a left-exact localization of a presheaf $\infty$–category. In such a context, colimits are computed as homotopy colimits by construction (see Lurie, Higher Topos Theory). Therefore, any pushout diagram is automatically a homotopy pushout. In particular, the colimit $A \cup B$ computed in $\mathcal{E}$ is the homotopy colimit of the span. Restricting to the sub-$\infty$–category $\mathcal{O}$ of SpherePop configurations, we obtain the claimed equivalence as long as $\mathcal{O}$ is closed under such pushouts. $\square$

*Remark* 24.3. This theorem complements the model-categorical argument given earlier. The $\infty$–topos viewpoint is conceptually cleaner: SpherePop merges are simply colimits in an $\infty$–category of configurations, and hence automatically homotopy colimits.

# 25 Entropic Potential as Shifted Symplectic Form

Finally, we sketch how entropic structure may enter as a (shifted) symplectic form in the PTVV sense, bridging lamphrodynamic entropy and derived symplectic geometry.

Let $\mathcal{M}$ be a derived stack parameterizing RSVP fields $(\Phi, v, S)$ on a space $X$, and consider the cotangent complex $\mathbb{L}_{\mathcal{M}}$. A shifted symplectic structure is given by a closed, nondegenerate 2-form

$$\omega \in \mathcal{A}^{2,cl}(\mathcal{M})[n]$$

for some shift $n$.

**Definition 25.1** (Entropic Symplectic Form)**.** Suppose there exists a pairing

$$\langle \cdot, \cdot \rangle_S : \mathbb{T}_{\mathcal{M}} \otimes \mathbb{T}_{\mathcal{M}} \to \mathcal{O}_{\mathcal{M}}[n]$$

induced by an entropy field $S$, where $\mathbb{T}_{\mathcal{M}}$ is the tangent complex. The corresponding 2-form $\omega_S$ is called an *entropic symplectic form* if it is:

1. skew-symmetric up to homotopy,
2. non-degenerate in the derived sense,
3. closed with respect to the de Rham differential.

Intuitively, $\omega_S$ measures "entropic curvature" in field space: it pairs lamphrodynamic modes with their conjugate variations in a way that depends explicitly on the entropy landscape.

**Conjecture 25.2** (PTVV-type Entropic Symplectic Structure)**.** *There exists a choice of derived moduli stack $\mathcal{M}_{\text{RSVP}}$ of lamphrodynamic fields and an integer shift $n$ such that:*

1. *$\mathcal{M}_{\text{RSVP}}$ is equipped with an $n$–shifted symplectic form $\omega_S$ encoding entropy descent;*
2. *the associated Hamiltonian function $\Theta$ reproduces the RSVP lamphrodynamic equations as Hamiltonian vector fields on $T^*[n]\mathcal{M}_{\text{RSVP}}$.*

If realized, this conjecture would place RSVP squarely within the PTVV landscape of shifted symplectic geometry, providing a rigorous bridge between entropic physics and derived symplectic structures.

**Part VII**

# Conclusion, Future Work, and Conjectural Foundations

## 26 General Conclusions

The research projects surveyed in this essay represent a wide range of interdependent investigations into lamphrodynamic physics, derived geometry, semantic computation, and cosmological interpretation. The Relativistic Scalar–Vector Plenum serves as the common ontological core, while DAG, shifted symplectic geometry, L–systems, spectral analysis, and amplitwistor frameworks offer mathematical lenses through which lamphrodynamic phenomena can be formalized or quantified.

Although most results developed so far remain conjectural, the gradual emergence of a unified research landscape suggests that many of these domains are not merely analogous but potentially equivalent in the sense of sharing a common derived geometric foundation.

## 27 Programmatic Future Work

Future investigations may be grouped under several large objectives:

1. **Formalization:** derive rigorous action functionals for RSVP, along with their Euler–Lagrange equations and boundary conditions.
2. **Quantization:** apply AKSZ/BV methodology to quantize lamphrodynamic fields and propose a coherent theory of quantum RSVP.
3. **Cosmological Validation:** numerically simulate redshift and lensing under entropic curvature, comparing predictions with observational data sets.
4. **Spectral Neuroscience:** examine whether entropic and lamphrodynamic ideas shed light on cortical resonance, consciousness, and neural complexity.
5. **Semantic Implementation:** unify SpherePop, DAG–RSVP, and semantic infrastructures into a general formalism of entropy–respecting symbolic and geometric computation.

## 28 Candidate Theorems and Conjectures

The following conjectures are not proven, but may serve as a foundation for future work.

[Entropy–Curvature Correspondence] Let $(M, g, \Phi, v, S)$ be an RSVP configuration satisfying lamphrodynamic field equations. Then gravitational curvature $R_{ab}$ admits a decomposition

$$R_{ab} = R_{ab}^{\text{matter}} + R_{ab}^{\Lambda}$$

where $R_{ab}^{\Lambda}$ is algebraically equivalent to an entropy–weighted tensor determined by $(\nabla S, v)$ up to gauge transformation.

[Derived Sigma–Model Equivalence] Lamphrodynamic field evolution can be equivalently described as a derived sigma model on a mapping space of rewrite rules, i.e.,

$$\text{RSVP} \simeq \sigma\text{–model}(\mathcal{R})$$

in a suitably defined derived category.

[Spectral Phase Transition] For appropriate boundary conditions, RSVP admits entropy–weighted spectral phase transitions whose eigenmodes capture semantic or cognitive structures within a lamphrodynamic manifold.

# Part VIII

# Supplementary Lemmas and Proof Sketches

## 29 SpherePop Gradient Flow and Energy Dissipation

In this section we provide a basic variational argument showing that the SpherePop collapse operation can be interpreted as a gradient flow for an energy functional, at least in a simplified continuum limit.

Recall the energy functional from the SpherePop section:

$$\mathcal{A}[\sigma] = \int_X \Big(\alpha \|\nabla\sigma\|^2 + \beta S(\sigma)\Big)\, dV, \tag{11}$$

where $\sigma$ is a configuration of spherical populations on a manifold $X$ and $S(\sigma)$ is a scalar entropy density.

**Proposition 29.1** (Gradient Flow Energy Dissipation)**.** *Suppose $\sigma(t, x)$ evolves according to the gradient flow*

$$\partial_t \sigma = -\frac{\delta \mathcal{A}}{\delta \sigma}, \tag{12}$$

*with appropriate boundary conditions (e.g. $\sigma$ compactly supported or $\nabla\sigma \cdot n = 0$ on $\partial X$). Then*

$$\frac{d}{dt}\mathcal{A}[\sigma(t)] \le 0$$

*for all t for which the flow is defined. In particular, $\mathcal{A}$ is nonincreasing along solutions, and strictly decreasing unless $\delta\mathcal{A}/\delta\sigma = 0$.*

*Proof.* We compute the time derivative formally:

$$\frac{d}{dt}\mathcal{A}[\sigma(t)] = \int_X \left\langle \frac{\delta \mathcal{A}}{\delta \sigma}, \partial_t \sigma \right\rangle dV.$$

By the definition of the gradient flow (12), we have $\partial_t \sigma = -\delta\mathcal{A}/\delta\sigma$. Substituting,

$$\frac{d}{dt}\mathcal{A}[\sigma(t)] = -\int_X \left\| \frac{\delta \mathcal{A}}{\delta \sigma} \right\|^2 dV \le 0.$$

The integral is nonpositive and vanishes if and only if the integrand is zero almost everywhere, i.e. when $\delta\mathcal{A}/\delta\sigma = 0$, which means $\sigma$ is a critical point of $\mathcal{A}$. This proves the claim. □

*Remark* 29.2. This proposition shows that, in a continuum approximation, the collapse operator can be modeled as a gradient flow on configuration space. In a discrete SpherePop implementation (finite collection of merging spheres), one expects an analogous monotonicity of $\mathcal{A}$ under merge–collapse moves, up to discretization error.

# 30 Spectral Geometry: Entropy-Weighted Laplacian

We now consider a basic spectral result justifying the use of entropy-weighted spectral measures on compact manifolds.

Let $(M, g)$ be a compact Riemannian manifold without boundary and let $w$ be a smooth positive weight function on $M$. Define the weighted Laplacian

$$\Delta_w f = -\frac{1}{w}\nabla_a\Big(w \nabla^a f\Big),$$

acting on smooth functions $f \in C^\infty(M)$.

**Theorem 30.1** (Discrete Spectrum of the Weighted Laplacian)**.** *On a compact Riemannian manifold $(M, g)$ with smooth strictly positive weight $w$, the operator $\Delta_w$ admits a discrete* *spectrum*

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \uparrow \infty,$$

*with corresponding eigenfunctions* $\{\psi_k\}_{k=0}^{\infty}$ *forming a complete orthonormal basis of* $L^2(M, w\, dV_g)$.

*Proof (sketch).* Define the weighted inner product

$$\langle f, h \rangle_w = \int_M f h\, w\, dV_g.$$

Consider the quadratic form

$$Q[f] = \int_M \|\nabla f\|^2\, w\, dV_g,$$

defined on $H^1(M)$ with respect to this inner product. Standard elliptic theory implies that $Q$ is densely defined, closed, and nonnegative. The representation theorem for self-adjoint operators then yields a unique nonnegative self-adjoint operator $A$ associated to $Q$ such that

$$Q[f] = \langle Af, f \rangle_w$$

for $f$ in the domain of $A$. One checks that $A$ agrees with $\Delta_w$ on $C^\infty(M)$.

Since $M$ is compact and the embedding $H^1(M) \hookrightarrow L^2(M)$ is compact, the resolvent $(A + \alpha I)^{-1}$ is compact for any $\alpha > 0$. It follows that $A$ has purely discrete spectrum with finite-dimensional eigenspaces and that eigenfunctions form an orthonormal basis. The ordering and divergence of eigenvalues follow from standard variational principles (Min–Max theorem) applied to $Q$. Details can be found in any standard text on elliptic operators on compact manifolds. □

*Remark* 30.2. In the RSVP context, one may select weights $w = e^{-S}$ or similar, where $S$ is an entropy density, so that high-entropy regions are suppressed spectrally. The theorem guarantees that the resulting entropy-weighted spectrum remains discrete and well-behaved.

# 31 Closing Remarks

The overarching hypothesis of RSVP is that lamphrodynamics offers a single, unifying language for gravity, cosmology, agency, semantics, and cognition. While speculative, the framework presents a coherent program of theoretical research that draws from established mathematics (derived geometry, homotopical methods, spectral theory) in a way that anticipates novel interpretations of physical and cognitive phenomena.

## Acknowledgements

The author thanks ongoing collaborators, colleagues, and correspondents for feedback, inspiration, and many valuable discussions on RSVP and its associated theoretical frameworks.

# Part IX

# Open Problems and Research Directions

## A RSVP Field Theory

### A.1 Analytic Questions

1. (**Nonlinear well-posedness**) Establish local (or global) well-posedness of the full lamphrodynamic PDE system containing nonlinear advection, torsion, and entropy-coupled potentials. Current results cover only linear or weakly nonlinear regimes.

2. (**A priori estimates**) Derive a priori energy bounds that remain valid beyond perturbative settings, e.g. upper bounds controlling blow-up or singular torsional behavior in curvature-dominated regimes.

3. (**Torsion–entropy/metric coupling**) Provide conditions under which torsion terms $\mathcal{T}$ preserve hyperbolicity or parabolic regularization; determine regimes where torsion dominates entropy descent and potentially induces lamphrodynamic shock-like phenomena.

### A.2 Geometric/Variational Questions

1. (**Variational formulation**) Give a rigorous derivation of RSVP field equations from a variational principle on a derived manifold (or stack) including boundary conditions, natural lamphrodynamic momenta, and symplectic form.

2. (**Shifted symplecticity**) Determine the shifted degree of the RSVP symplectic structure within the PTVV classification, ideally identifying it with a natural AKSZ degree.

3. (**Quantization**) Construct a BV quantization of RSVP using shifted cotangent stacks and verify the classical master equation for a lamphrodynamic AKSZ action.

# B SpherePop Calculus and Derived Geometry

## B.1 Model Category / $\infty$-Topos Questions

1. (**Model structure**) Equip SpherePop configurations with a model category (or $\infty$–category) structure in which merge and collapse are admissible cofibrations/weak equivalences respecting derived intersections.

2. (**Homotopy colimits**) Prove that iterated merge operations compute derived (homotopy) colimits in the candidate $\infty$–category, ensuring a compositional semantics for SpherePop computation.

3. (**Collapse vs truncation**) Formally prove that collapse is equivalent to a suitable truncation or localization of $\mathbb{R}\mathcal{M}_{\mathrm{SP}}$ preserving boundary strata.

## B.2 Derived Moduli and Symplectic Structure

1. (**Derived enhancement**) Construct the derived moduli stack $\mathbb{R}\mathcal{M}_{\mathrm{SP}}$ of SpherePop configurations satisfying natural stability and finiteness conditions.

2. (**PTVV compatibility**) Exhibit an entropic (shifted) symplectic 2-form $\omega_{\mathrm{SP}}$ as a closed, nondegenerate element of $\mathcal{A}^{2,cl}$ on $T^*[n](\mathbb{R}\mathcal{M}_{\mathrm{SP}})$.

3. (**Universal property**) Identify universal mapping properties of $\kappa$ (collapse) as a final object among shape-preserving maps that factor through boundary semantics.

# C Cosmology and Gravity as Entropy Descent

1. (**Cosmological redshift**) Derive entropic redshift formulas from RSVP field equations and compare with standard $\Lambda$CDM predictions for CMB power spectra and BAO measurements.

2. (**Lensing and structure formation**) Develop lamphrodynamic lensing equations incorporating entropy curvature and torsion terms; compare numerical simulations with observed cluster lensing and large-scale structure statistics.

3. (**Strong-field regimes**) Determine whether lamphrodynamic entropy curvature can mimic black-hole metrics or singularity formation in general relativity; identify regimes where lamphrodynamic torsion produces new strong-field phenomena.

## D Spectral Geometry, Cymatics, and Neural Resonance

1. (**Spectral phase transitions**) Establish existence of entropy-weighted spectral phase transitions for RSVP operators and characterize the resulting eigenmodes.

2. (**Boundary conditions**) Analyze effects of cortical boundary geometry on lamphrodynamic resonance modes using entropy-weighted Laplacians and derived boundary conditions.

3. (**Functional imaging**) Test theoretical predictions against ultrafast fMRI or ECoG data, seeking evidence of lamphrodynamic resonance or entropy-driven spectral signatures.

## E Amplitwistors and Scattering Geometry

1. (**Amplitwistor correspondence**) Formulate a derived ambitwistor correspondence for RSVP fields, identifying canonical maps from lamphrodynamic phase space to ambitwistor space.

2. (**Factorization**) Determine factorization rules for RSVP scattering states and clarify whether entropic flows admit twistor-style factorization of interactions.

3. (**Quantization pathway**) Provide an AKSZ/BV route from derived stacks of lamphrodynamic data to quantum scattering amplitudes, paralleling modern twistor/ambitwistor theory.

## F Narrative, Creative, and Aesthetic Frameworks

1. (**Narrative formalization**) Develop a precise theory of narrative structure that encodes semantic recursion, lamphrodynamic dynamics, and cognitive curvature in a geometric or homotopical language.

2. (**Cinematic semantics**) Relate visual composition and soundtrack modulation to RSVP spectral invariants, suggesting aesthetic rules constrained by entropy descent.

3. (**Ethical phenomenology**) Explore ethical interpretations of collapse, merge, and entropic smoothing as narrative metaphors for agency, responsibility, and recursive emergence.

## Summary

The open problems listed here illustrate that many aspects of RSVP theory, SpherePop geometry, lamphrodynamic cosmology, and derived semantics remain formally conjectural. Nonetheless, each domain now forms part of a coherent research ecosystem that invites rigorous mathematical development, empirical testing, and creative interpretation in an integrated framework.

# Part X

# Glossary of Symbols and Notation

This appendix records the most frequently used symbols and notational conventions across RSVP theory, lamphrodynamics, SpherePop calculus, and derived-geometric formalization. Items are grouped thematically; some may be redefined in specific chapters with additional hypotheses.

## A Geometric and Differential Operators

[leftmargin=2.5cm,style=nextline]A smooth Lorentzian or Riemannian manifold with metric $g$. Metric-compatible covariant derivative (the Levi–Civita connection unless otherwise specified). Laplace–Beltrami operator associated to $g$ (sign conventions vary; here $\Delta$ is nonpositive in Riemannian signature). Scalar curvature and Ricci tensor of $(M, g)$. Stress–energy tensor (may include lamphrodynamic or entropic sources). Lie derivative with respect to a vector field $X$. Exterior derivative acting on differential forms.

## B RSVP Lamphrodynamic Fields

[leftmargin=2.5cm,style=nextline]Scalar lamphrodynamic potential describing negentropic tendency. Vector lamphrodynamic flow field (often subject to divergence or torsion constraints). Entropy field coupling geometric and lamphrodynamic degrees of freedom. Torsion tensor (when not the stress tensor); used for lamphrodynamic torsion. Effective cosmological constant induced by entropy curvature. General lamphrodynamic modification to Einstein equations. Coupling constants in lamphrodynamic Lagrangians or potentials.

## C SpherePop Calculus and Semantic Geometry

[leftmargin=2.5cm,style=nextline]SpherePop configuration or density field; may denote a collection of regions $\{B_i\}$ or a scalar field approximation. Potential term coupling SpherePop density and entropy. Collapse operator mapping configurations to boundary-equivalent representatives. Intersection of regions in a SpherePop configuration. Merge configuration (pushout or homotopy colimit in categorical formalizations).

Homotopy colimit in model-categorical or $\infty$–categorical settings. Ordinary categorical colimit.

# D Entropy, Spectral Geometry, and Resonance

[leftmargin=2.5cm,style=nextline]Positive weight function, often $w = e^{-S}$ in entropy-weighted geometry. Weighted Laplacian $\Delta_w f := -\frac{1}{w}\nabla_a(w\nabla^a f)$. Eigenfunctions and eigenvalues of Laplacian or entropy-weighted operators. Angular frequency; also denotes symplectic form (context-dependent, indicated locally). Coupling constants in entropy/lamphrodynamic PDEs.

# E Derived Geometry and Shifted Structures

[leftmargin=2.5cm,style=nextline]A derived moduli stack indexing field configurations or geometric objects. Derived enhancement of a moduli problem.

# F AKSZ / BV Formalism

[leftmargin=2.5cm,style=nextline]Source dg-manifold (often encoding spacetime plus ghost/antifield degrees). Target derived manifold or stack for AKSZ construction (field space). AKSZ action functional, satisfying classical master equation. Poisson bracket (shifted, degree determined by context); used for master equation. BV differential encoding gauge symmetries and lamphrodynamic redundancies.

# G Cosmology and Gravitational Quantities

[leftmargin=2.5cm,style=nextline]Spacetime metric with signature $(-,+,+,+)$ unless stated otherwise. Hubble parameter as a function of redshift $z$. Luminosity and angular diameter distances. Spatial curvature parameter in cosmology (not to be confused with shift degree). Energy density and pressure in effective fluid descriptions.

# H Miscellaneous and Contextual Symbols

[leftmargin=2.5cm,style=nextline]Ambient manifold (physical, semantic, or computational). Test space used in moduli functor or derived mapping stack. Volume form induced by metric $g$. Generic small parameters used in energy estimates and perturbative arguments. Generic positive constants (energy estimates, inequalities), values may vary in different statements.

## Notational Conventions

[illegible]e components; summation index convention is always assumed.

- Bold symbols $\boldsymbol{v}$ are sometimes used to emphasize spatial vector fields; context determines whether $v$ is spacetime or purely spatial.
- Functional derivatives $\delta/\delta\Phi$ are formal unless explicitly defined via variational calculus in a geometric (derived) setting.
- $\simeq$ denotes equivalence in homotopy category or derived category; $\cong$ denotes strict isomorphism in the underlying category.

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